Model-based RL

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CS 1840 Fall 2026

Announcements

HW3 ~~will be~~ released ~~on~~ Friday!

Midterm will be in 2 weeks on Mon 10/26!

Check the course website for Office Hours:
https://harvard-cs-1840.github.io/2026/

Finishing Up From Last Lecture



1. Q Learning

2. Revisit TD: Off-policy TD Learning

TD Learning

Given (s, a, r, s'), where $a \sim \pi(\cdot \mid s), s' \sim P(\cdot \mid s, a)$, TD updates:

$$\hat{V}^{\pi}(s) \Leftarrow \hat{V}^{\pi}(s) + \eta \left(r + \gamma \hat{V}^{\pi}(s') - \hat{V}^{\pi}(s') \right)$$

↑ ↑ $r + \gamma \hat{V}^{\pi}(s')$ – target

On-policy: data is generated from the policy π itself

Off-policy: data is generated from policy π_b where $\pi_b \neq \pi$

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Q: is Q-learning off-policy or on-policy?

Motivation for off-policy evaluation

Counterfactual: what would happen if I did something different?

Off-policy TD Learning

Setting: data is generated by π_b, but we want to estimate V^π for some $\pi \neq \pi_b$

Key trick: importance weighting

$$V^\pi(s) = \mathbb{E}_{a\sim\pi(\cdot|s)}\left(r(s,a) + \gamma \mathbb{E}_{s'\sim P(\cdot|s,a)} V^\pi(s')\right)$$

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$$= \mathbb{E}_{a\sim\pi_b(\cdot|s)} \frac{\pi(a\,|\,s)}{\pi_b(a\,|\,s)}\left(r(s,a) + \gamma \mathbb{E}_{s'\sim P(\cdot|s,a)} V^\pi(s')\right)$$

$\sum \pi_b(a|s) \cdot \frac{\pi(a|s)}{\pi_b(a|s)} [\ \ldots\ )$

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Now action is sampled from π_b

Importance weight

Off-policy TD Learning

Given (s, a, r, s'), where $a \sim \pi_b(\cdot \mid s), s' \sim P(\cdot \mid s, a)$,

Off-policy TD updates as follows:

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Case 1: $\pi(a \mid s)$ is large but $\pi_b(a \mid s)$ is small

Case 2: $\pi(a \mid s)$ is small but $\pi_b(a \mid s)$ is large

Recap: Planning algorithm for computing

We assumed that $P(s' \mid s, a), r(s, a) \forall s, a, s'$ are **known**

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Value Iteration:

$$Q^{t+1}(s, a) \Leftarrow r(s, a) + \mathbb{E}_{s' \sim P(\cdot|s,a)} \max_{a'} Q^t(s', a'), \forall s, a$$

compute exactly!

$r(s,a) + \sum_{s'} P(s'|s,a) \cdot \max_{a'} Q^t(s', a')$ Yum

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Policy Iteration:

$$\pi^{t+1}(s) = \arg\max_a Q^{\pi^t}(s, a), \text{ for all } s$$

$Q^{\pi_{t+1}}(s,a) \geq Q^{\pi_t}(s,a) \quad \forall (s,a)$

$\pi^*(a|s) = \arg\max_a Q^*(s,a) \quad \forall s$

Recap: Value-based Learning

When $P(s' \mid s, a)$ is unknown, Q-learning aims to learn $Q^\star$ directly

$$\hat{Q}(s,a) \Leftarrow \hat{Q}(s,a) + \left(r(s,a) + \max_{a'} \hat{Q}(s',a') - \hat{Q}(s,a) \right)$$

Q-target

where $a \sim \epsilon$-greedy$(\hat{Q})$, and $s' \sim P(\cdot \mid s, a)$, $r = r(s,a)$

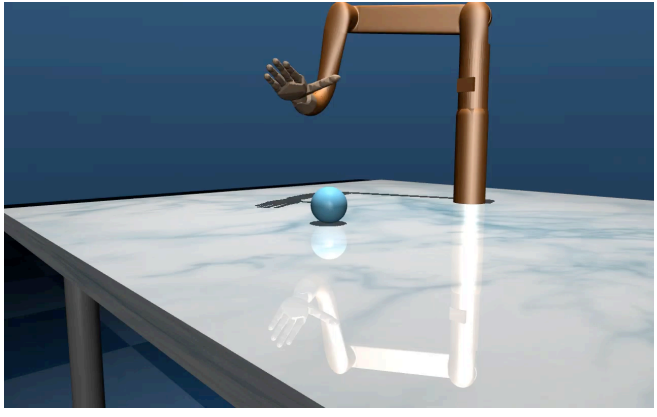
w/ prob ϵ select uniform action

w/ prob $1-\epsilon$, select action argmax $\vec{Q}[s,a]$

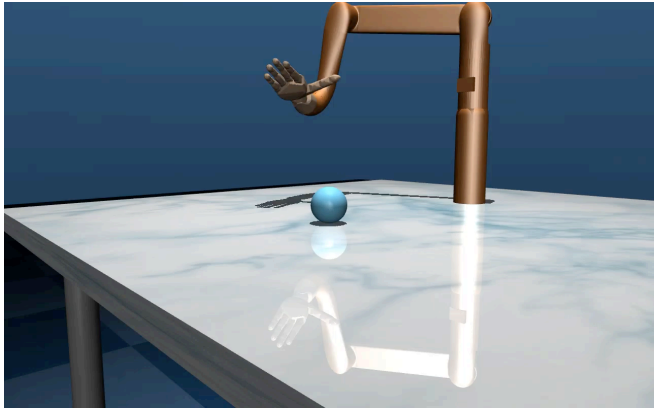
Questions for Today:

Can we **learn the transition** from data and then compute its optimal policy;
and what performance guarantee we can get?

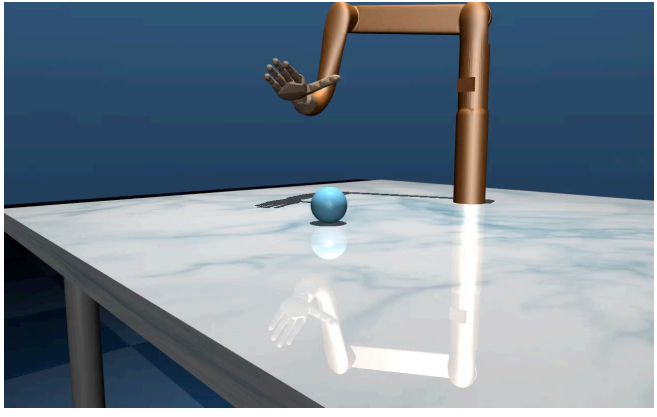
Motivation for Model-based Approach



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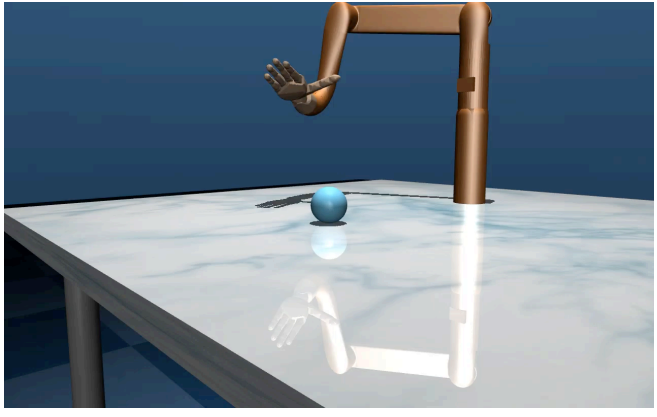


While we cannot model the exact analytical dynamics,

we can learn it from data $\{s, a, s'\}$

Denote $\hat{T}$ as the learned model

Motivation for Model-based Approach



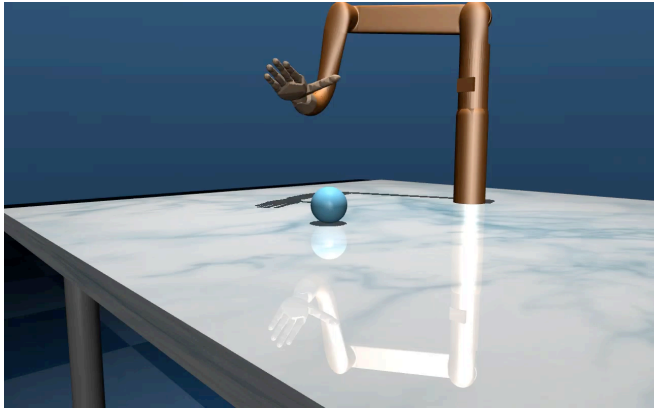
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$\widehat{\pi}^{\star} = \mathrm{VI}(\widehat{P}, r)$

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MDP(S, A, P, r, γ)

Then we do planning: e.g.,

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Learn $\hat{P}$

(Often in practice we iterate the above process)

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Potential benefits of learning model over Q-learning

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There are cases where model is much easier to learn than value function

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Potential benefits of learning model over Q-learning

There are cases where model is much easier to learn than value function

Once model is learned, we can optimize different rewards (i.e., multi-task)

Outline:

$\hat{P} \approx P$

1. Simulation lemma:
What is the performance of π under $(\widehat{P}, r)$

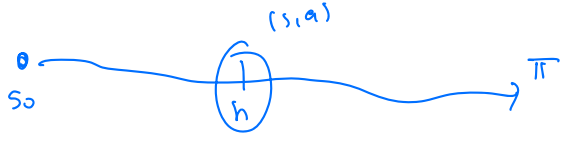
2. **Algorithm:** estimate $\widehat{P}$ from data
and compute $\widehat{\pi}^\star$—the optimal policy of $\widehat{P}$

3. **Analyzing** the performance $\widehat{\pi}^\star$ under (P, r)

State-action distribution

Given a policy π and s_0, we denote $\mathbb{P}_h^\pi(s, a \mid s_0)$ as the **prob of reaching (s, a) at time h, given we start at s_0 at $h = 0$**

$\mathbb{P}_h^\pi(s, a \mid s_0) = \mathbb{P}_\pi(S_h = s,\ a_h = a \mid s_0, \pi)$



$\sum_{s,a} \mathbb{P}_h^\pi(s, a \mid s_0) = 1$

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Denote $d_{s_0}^\pi$ as the average state-action distribution:

$$d_{s_0}^\pi(s, a) = (1-\gamma) \sum_{h=0}^{\infty} \gamma^h \mathbb{P}_h^\pi(s, a \mid s_0)$$

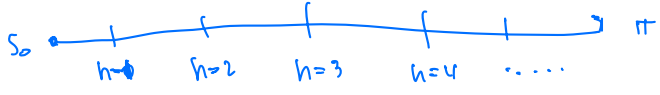
$$\sum_{s,a} \mathbb{P}_h^\pi(s,a) = 1$$

$$= \sum_{h=0}^{\infty} \gamma^h \sum_{s,a} \mathbb{P}_h^\pi$$

$$= \sum_{h=0}^{\infty} \gamma^h \sum_{s,a} \mathbb{P}_h^\pi \quad \sum_a$$

$$= \sum_{h=0}^{\infty} \gamma^h = \frac{1}{1-\gamma}$$

$$(1-\gamma)\left(\gamma^0 \mathbb{P}_0^\pi(s,a) + \gamma^1 \mathbb{P}_1^\pi(s,a) + \gamma^2 \mathbb{P}_2^\pi(s,a)\right)$$



A key fundamental question in Model-based RL:

Given two transitions $\hat{P}$ and P, how would π behave differently in $\hat{P}$ and P?

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Denote:

$$\widehat{V}^{\pi}(s_0) = \mathbb{E}\left[\sum_{h=0}^{\infty} \gamma^h r(s_h, a_h) \,|\, \pi, \widehat{P}\right]; \quad V^{\pi}(s_0) = \mathbb{E}\left[\sum_{h=0}^{\infty} \gamma^h r(s_h, a_h) \,|\, \pi, P\right];$$

A key fundamental question in Model-based RL:

Given two transitions $\hat{P}$ and P, how would π behave differently in $\hat{P}$ and P?

$\|\hat{V}^{\pi} - V^{\pi}\| \leq ??$

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What is the difference between $\widehat{V}^{\pi}(s_0)$ & $V^{\pi}(s_0)$?

In other words, how does the model error propagate to values

Simulation Lemma:

$$\widehat{V}^{\pi}(s_0) - V^{\pi}(s_0) = \frac{\gamma}{1-\gamma}\mathbb{E}_{s,a\sim d^{\pi}_{s_0}}\left[\mathbb{E}_{s'\sim \widehat{P}(s,a)}\widehat{V}^{\pi}(s') - \mathbb{E}_{s'\sim P(s,a)}\widehat{V}^{\pi}(s')\right]$$

π under true model

$$\leq \frac{\gamma}{(1-\gamma)^2}\mathbb{E}_{s,a\sim d^{\pi}_{s_0}}\left\| \widehat{P}(\cdot\,|\,s,a) - P(\cdot\,|\,s,a)\right\|_1$$

$$\mathbb{E}_{s'\sim\hat{P}(s,a)}\left[\hat{V}^{\pi}(s')\right] - \mathbb{E}_{s'\sim P(s,a)}\left[\hat{V}^{\pi}(s')\right]$$

$$= \sum_{s'}\left[\hat{P}(s'\,|\,s,a) - P(s'\,|\,s,a)\right]\hat{V}^{\pi}(s')$$

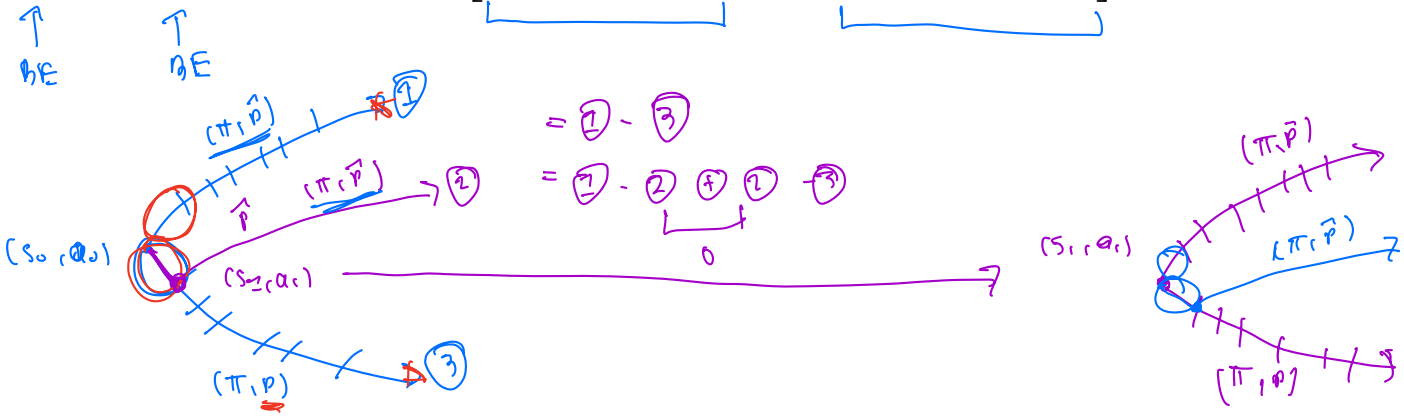
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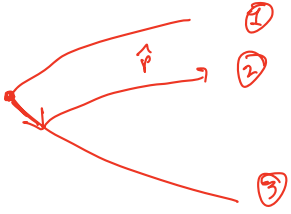
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1 2 3



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$$+\gamma\mathbb{E}_{a_0\sim\pi(\cdot|s_0),s_1\sim P(s_0,a_0)}\left[\widehat{V}^{\pi}(s_1) - V^{\pi}(s_1)\right] \quad (2) - (3)$$

Summary so far:

$$\widehat{V}^{\pi}(s_0) - V^{\pi}(s_0) = \frac{\gamma}{1-\gamma}\mathbb{E}_{s,a\sim d^{\pi}_{s_0}}\left[\mathbb{E}_{s'\sim \widehat{P}(s,a)}\widehat{V}^{\pi}(s') - \mathbb{E}_{s'\sim P(s,a)}\widehat{V}^{\pi}(s')\right]$$

$$\leq \frac{\gamma}{(1-\gamma)^2}\mathbb{E}_{s,a\sim d^{\pi}_{s_0}}\left\| \widehat{P}(\cdot\,|\,s,a) - P(\cdot\,|\,s,a)\right\|_1$$

$|\mathbb{E}_{x\sim p} f(x) - \mathbb{E}_{x\sim p} g(x)| \leq \|f(x)\|_\infty \|p-q\|_1$

$r \in [0,1]$

$V^{\pi} \in [0, \frac{1}{1-\gamma}]$

$$\left|\sum_{s'}\left[\widehat{P}(s'|s,a) - P(s'|s,a)\right] V^{\pi}(s')\right| \leq \|V^{\pi}(s')\|_\infty \left\|\widehat{P}(s'|s,a) - P(s'|s,a)\right\|_1$$

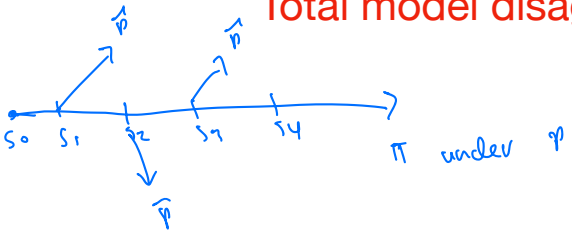
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$$\widehat{V}^{\pi}(s_0) - V^{\pi}(s_0) = \frac{\gamma}{1-\gamma}\mathbb{E}_{s,a\sim d^{\pi}_{s_0}}\left[\mathbb{E}_{s'\sim \widehat{P}(s,a)}\widehat{V}^{\pi}(s') - \mathbb{E}_{s'\sim P(s,a)}\widehat{V}^{\pi}(s')\right]$$

$$\leq \frac{\gamma}{(1-\gamma)^2}\mathbb{E}_{s,a\sim d^{\pi}_{s_0}}\left\| \widehat{P}(\cdot\,|\,s,a) - P(\cdot\,|\,s,a)\right\|_1$$

Total model disagreement over the real traces



Outline:



1. Simulation lemma:

What is the performance of π under any estimator $\widehat{P}$

2. **Algorithm:** estimate ($\widehat{P}$, $\widehat{r}$) from data
and compute $\widehat{\pi}^{\star}$ —the optimal policy of ($\widehat{P}$, $\widehat{r}$)

3. **Analyzing** the performance $\widehat{\pi}^{\star}$ under (P, r)

A Model-based Algorithm

Assume reward r is known (just for analysis simplicity):

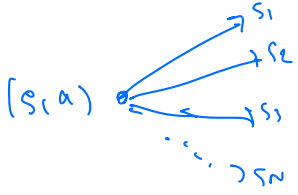
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1. Model fitting:

$\forall s, a$: collect N next states, $s_i' \sim P(\cdot \mid s, a), i \in [N]$; set

$$\widehat{P}(s' \mid s, a) = \frac{\sum_{i=1}^{N} \mathbf{1}\{s_i' = s'\}}{N};$$



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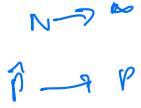
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2. Planning w/ the learned model:

$$\widehat{\pi}^{\star} = \textbf{PI}\left(\widehat{P}, r\right)$$



Detour: estimating mean of Bernoulli distribution

Given: we have a biased coin:

With probability p, it gives +1, and w/ prob 1-p, it gives -1;

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$$\hat{p} = \frac{\sum_{i=1}^{N} \mathbf{1}\{x_i = +1\}}{N}$$

w/ probability at least $1 - \delta$ we will have $|\hat{p} - p| \leq O\left(\sqrt{\frac{\ln(1/\delta)}{N}}\right)$

(proof out of scope)

Model-based RL

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Estimating the prob of observing s' at (s, a)

2. Planning w/ the learned model:

$$\widehat{\pi}^\star = \textbf{PI}\left(\widehat{P}, r\right)$$

Model-based RL

1. Model fitting:

$\forall s, a$: collect N next states,

$s_i' \sim P(\cdot \mid s, a), i \in [N]$; set

$$\widehat{P}(s' \mid s, a) = \frac{\sum_{i=1}^{N} \mathbf{1}\{s_i' = s'\}}{N};$$

Estimating the prob of observing s' at (s, a)

We can show:

$$\|\hat{P}(\cdot \mid s, a) - P(\cdot \mid s, a)\|_1 \lesssim \sqrt{\ln(1/\delta)/N}, \forall s, a$$

2. Planning w/ the learned model:

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Summary so far:

By collecting data (next state) at very (s, a), we build an estimator $\hat{P}$ that is close to P

(e.g., possible to show error shrinks $\approx 1/\sqrt{N}$)

Outline:

✓ **1. Simulation lemma:**
What is the performance of π under any estimator $\widehat{P}$

✓ 2. **Algorithm:** estimate $(\widehat{P}, \widehat{r})$ from data
and compute $\widehat{\pi}^\star$—the optimal policy of $(\widehat{P}, \widehat{r})$

3. **Analyzing** the performance $\widehat{\pi}^\star$ under (P, r)

Performance of the learned policy

Now assume that $\hat{P}$ is nearly accurate, i.e., $\|\hat{P}(\cdot \mid s,a) - P(\cdot \mid s,a)\|_1 \leq \epsilon, \forall s,a$

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$$\leq V^\star(s_0) - \widehat{V}^{\pi^\star}(s_0) + \widehat{V}^{\hat{\pi}^\star}(s_0) - V^{\hat{\pi}^\star}(s_0)$$

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$$\le \frac{2\gamma}{(1-\gamma)^2}\cdot\epsilon,$$

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1. A model-based RL **Algorithm for small-size MDP**

$$\widehat{P}(s' \mid s, a) = \frac{\sum_{i=1}^{N} \mathbf{1}\{s_i' = s'\}}{N}, \forall s, a; \quad \widehat{\pi}^\star = \mathbf{PI}\left(\widehat{P}, r\right)$$

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3. Good model leads to a good policy, up to some error amplification based on effective horizon

Feedback and Attendance

https://bit.ly/4gYn2nC

